Uses

Chrysler Building, New York City, USA

Stainless steel is quite resistant to rain and wind.

Nuts and bolts

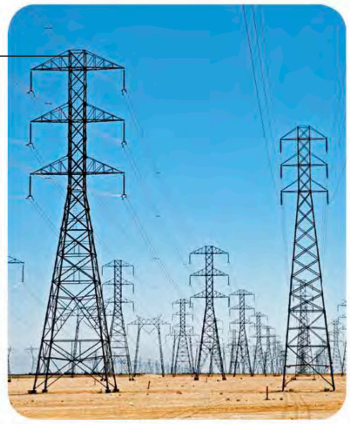
This fastener is made of strong steel.

Tractor

This steel body resists rusting.

Steel wool

Thin wires of steel are used to clean hard surfaces.

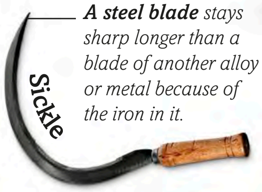


Transmission towers

These tall structures are made from stiff steel girders.

Cast iron pot

This iron pot retains heat well while cooking.



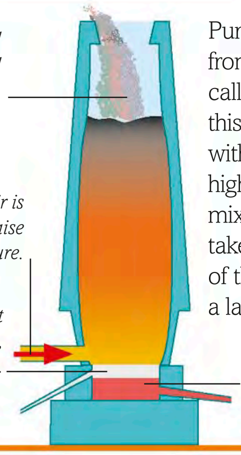
Sickle

A steel blade stays sharp longer than a blade of another alloy or metal because of the iron in it.

Iron filings and magnet

These small grains of pure iron are magnetic and are attracted to the end of a magnet.

SMELTING



1. Iron ore and coal are added to the furnace.

2. Hot air is added here to raise the temperature.

3. Impurities float on the pure metal, then released.

4. Pure iron sinks to the bottom, then removed.

Pure iron is separated from its ores in a process called smelting. During this process, iron reacts with carbon in coal at a high temperature. As the mixture burns, the carbon takes the impurities out of the ore, leaving behind a layer of pure iron.

our cells produce energy for the body to work). Foods containing iron include meats and green vegetables, such as **spinach**. When pure iron comes into contact with air and water, it develops a flaky, reddish-brown coating called rust, which weakens the metal. In order to make iron tougher, tiny amounts of carbon and other metals, such as nickel and titanium, are added to it. This forms an alloy called steel, which is used to make **bolts** and strong **tractor** bodies, among other applications. Adding the element chromium to steel creates a stronger alloy called stainless steel.